

Heisenberg's uncertainty principle for a partical in a box

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To verify that the particle in an infinite potential well obeys the Heisenberg Uncertainty Principle, we calculate the uncertainties in position and momentum from their expectation values. The normalized wavefunction is

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right), \quad 0 < x < a.$$

Hence,

$$|\psi_n(x)|^2 = \frac{2}{a} \sin^2\left(\frac{n\pi x}{a}\right).$$

The expectation value of the position is

$$\langle x \rangle = \int_0^a x |\psi(x)|^2 dx = \frac{2}{a} \int_0^a x \sin^2\left(\frac{n\pi x}{a}\right) dx.$$

Using

$$\int x \sin^2(\alpha x) dx = \frac{x^2}{4} - \frac{x \sin(2\alpha x)}{4\alpha} - \frac{\cos(2\alpha x)}{8\alpha^2},$$

and noting that

$$\sin(2n\pi) = 0, \quad \cos(2n\pi) = 1,$$

gives

$$\boxed{\langle x \rangle = \frac{a}{2}}.$$

Similarly,

$$\langle x^2 \rangle = \frac{2}{a} \int_0^a x^2 \sin^2\left(\frac{n\pi x}{a}\right) dx.$$

Using

$$\int x^2 \sin^2(\alpha x) dx = \frac{x^3}{6} - \frac{x \cos(2\alpha x)}{4\alpha^2} - \frac{(2\alpha^2 x^2 - 1) \sin(2\alpha x)}{8\alpha^3},$$

gives

$$\boxed{\langle x^2 \rangle = \frac{a^2}{3} - \frac{a^2}{2n^2\pi^2} = \frac{a^2}{3} \left(1 - \frac{3}{2n^2\pi^2}\right)}.$$

Therefore,

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$$

becomes

$$\Delta x = \sqrt{\frac{a^2}{3} - \frac{a^2}{2n^2\pi^2} - \frac{a^2}{4}}$$

which simplifies to

$$\boxed{\Delta x = \frac{a}{2n\pi} \sqrt{\frac{n^2\pi^2}{3} - 2}}.$$

For the particle in a box,

$$\langle p \rangle = 0,$$

and

$$\langle p^2 \rangle = \frac{n^2 \pi^2 \hbar^2}{a^2}.$$

Hence,

$$\Delta p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} = \boxed{\frac{n\pi\hbar}{a}}.$$

Multiplying the uncertainties,

$$\Delta x \Delta p = \frac{n\pi\hbar}{a} \cdot \frac{a}{2n\pi} \sqrt{\frac{n^2\pi^2}{3} - 2},$$

which simplifies to

$$\boxed{\Delta x \Delta p = \frac{\hbar}{2} \sqrt{\frac{n^2\pi^2}{3} - 2}}.$$

The smallest value occurs for the ground state ($n = 1$):

$$\Delta x \Delta p = \frac{\hbar}{2} \sqrt{\frac{\pi^2}{3} - 2} \approx 1.14 \frac{\hbar}{2}.$$

Since

$$1.14 > 1,$$

it follows that

$$\boxed{\Delta x \Delta p > \frac{\hbar}{2}},$$

which satisfies the Heisenberg Uncertainty Principle,

$$\boxed{\Delta x \Delta p \geq \frac{\hbar}{2}}.$$